

Course Code	:	MCS-230
Course Title	:	Digital Image Processing and Computer Vision
Assignment Number	:	MCAOL(IV)/230/Assign/2024-25
Maximum Marks	:	100
Weightage	:	30%
Last Dates for Submission	:	31st October, 2024 (For July, 2024 Session) 15th April, 2025 (For January, 2025 Session)

This assignment has sixteen questions of 5 Marks each, answer all questions. Rest 20 marks are for viva voce. You may use illustrations and diagrams to enhance the explanations. Please go through the guidelines regarding assignments given in the Programme Guide for the format of presentation.

- Q1:** What is image acquisition? Explain Optical, Analog and Digital image processing in brief. **(5 Marks)**
- Q2:** If the physical size of a medical image is 4×4 inches and the sampling resolution is 5 cycles/mm, then how many pixels per cycle are required to have a better-quality image? Will an image of size 512×512 be enough? **(5 Marks)**
- Q3:** Explain the types of Images based on (i) Attributes (ii) Based on Colour **(5 Marks)**
- Q4:** Solve the following problems: **(5 Marks)**
- What is the storage requirement for a 2024×2024 , 24-bit colour image?
 - Calculate pixel resolution of a camera in mega pixels, capturing an image of dimension: 3000×4000
 - Given an image is a gray scale image with aspect ratio of 8:2 and pixel resolution of 1000000 pixels, calculate the dimensions and the size of the image.
- Q5:** Explain how image enhancement is better in the frequency domain as compared to spatial domain. **(5 Marks)**
- Q6:** Explain the following Smoothing Filter(s): **(5 Marks)**
- Ideal Low Pass Filters (ILPF)
 - Butterworth Low Pass Filters (BLPF)
 - Gaussian Low Pass filters (GLPF)
- Q7:** Explain the following Image Sharpening Filter(s): **(5 Marks)**
- Ideal High Pass Filters (ILPF)
 - Butterworth High Pass Filters (BLPF)
 - Gaussian High Pass filters (GLPF)
- Q8:** Explain Mean Filters, and Median Filter with the help of a suitable example for each. **(5 Marks)**
- Q9:** Transform the RGB cube by its CMY cube. Label all the vertices. Also, interpret the colours at the edges with respect to saturation. **(5 Marks)**
- Q10:** Explain optical flow, in context of motion perception in computer vision. **(5 Marks)**
- Q11:** Explain epipolar geometry with the help of a suitable diagram in stereo vision system. **(5 Marks)**
- Q12:** What is camera calibration? Explain how it helps to estimate the intrinsic and extrinsic parameters of a camera. **(5 Marks)**

Q13: Explain K-means clustering methods with the help of a suitable example. Also, discuss the advantages and disadvantages of k -means clustering methods. **(5 Marks)**

Q14: Perform partitional clustering using Froggy's method for the data given in the table below with k-2 (two clusters). Use first two sample points (3,3) and (6,8) as seed points. **(5 Marks)**

S. No.	X	Y
1	3	3
2	6	8
3	10	10
4	4	4
5	6	6
6	14	12
7	20	18
8	22	20

Q15: Explain agglomerative hierarchical clustering and its types with the help of a suitable example. **(5 Marks)**

Q16: Explain Bayes classifier with the help of a suitable example. Also discuss its properties. **(5 Marks)**